

Proposed condensed test suite

Document: Proposed condensed SIM test suite
Prepared for: Review and approval
Date: 6 October 2026
Application: SIM Flight Testing checklist
Catalog: 77 Task / Element / Operation tests

77

Current catalog tests

18

Proposed missions

8

Single-path missions

REQUEST

Please approve a shorter official test list. The current SIM Flight Testing catalog has 77 Task / Element / Operation tests. Several of those tests can be flown as steps of one mission instead of as 77 separate flights. This document proposes grouping the 77 tests into 18 missions. It does not change the application until you approve a list.

RECOMMENDATION

Approve the 18-mission list below as the official condensed suite. If each corridor and viewpoint phase must still pass on its own, keep the original 77 tests and use missions only as flight order, not as a shorter catalog.

If you only want tests that truly fit one path with no repeats, approve missions 1, 2, 3 and 4, plus one resume each for loiter, POSCTL, EFL and Land. That is 8 missions and 18 original tests.

HOW TO READ THIS LIST

Missions 1-4 are real single flights. Score every listed step on that flight.

Missions 5-8 are the same interrupt at several phases. One flight can only use one phase. If each phase must pass on its own, do not collapse them.

Missions 11-18 are branches from the same state. One flight can only take one branch, so they are grouped by state, not by a single path.

PROPOSED MISSIONS

1. Corridor hive to hive

One flight. Score every step.

- Deployed on hive / disarmed -> arm
- Corridor / Take-off -> Complete
- Corridor / To First Waypoint -> Complete
- Corridor / Main/any mission corridor -> Complete
- Corridor / To HL -> Complete
- Corridor / Landing on hive -> Complete

2. Viewpoint swap to hive

One flight.

- Viewpoint / To hive from swap -> Complete

3. Return to Hive from viewpoint transit

One flight.

- Viewpoint / to viewpoint -> Return to Hive

4. RTL from viewpoint transit

One flight. Issue RTL, then confirm RTL completes.

- Viewpoint / to viewpoint -> RTL
- RTL -> Complete

5. In-flight loiter, then resume

Same operation at several phases. One flight can use only one phase; the others are repeats.

- Loiter from Take-off, To First Waypoint, Main/any mission corridor, To HL, to viewpoint, waiting for release, To hive from swap
- then Loiter -> mission

6. In-flight POSCTL, then resume

Same operation at several phases. One flight can use only one phase.

- POSCTL from the same in-flight elements as mission 5
- then PostCTL / In control -> mission

7. In-flight EFL, then resume

Same operation at several phases. One flight can use only one phase.

- EFL from the same in-flight elements as mission 5
- then EFL / Descending -> mission

8. In-flight Land / LIP, then resume

Same operation at several phases. Take-off has no Land case in the current suite.

- Land from To First Waypoint, Main/any mission corridor, To HL, to viewpoint, waiting for release, To hive from swap
- then LIP / Descending -> mission

9. POSCTL from the hive

Grounded / disarmed. Separate from arm.

- Deployed on hive / disarmed -> POSCTL

10. Arm from PostCTL

Requires the drone already in PostCTL.

- PostCTL / In control -> arm

11. From loiter, leave hold

Branches from the same state. One flight can take only one branch.

- Loiter -> POSCTL
- Loiter -> EFL
- Loiter -> Land

12. From PostCTL, leave hold

Branches from the same state. One flight can take only one branch.

- PostCTL / In control -> loiter
- PostCTL / In control -> EFL
- PostCTL / In control -> Land

13. LIP descent

Branches from LIP descending. One flight can take only one branch.

- LIP / Descending -> Complete
- LIP / Descending -> loiter
- LIP / Descending -> POSCTL
- LIP / Descending -> EFL
- LIP / Descending -> Land

14. After LIP landing

Branches from landed / disarmed. One flight can take only one branch.

- LIP / Landed (disarmed) -> arm
- LIP / Landed (disarmed) -> mission
- LIP / Landed (disarmed) -> POSCTL
- LIP / Landed (disarmed) -> EFL

15. LIP back to mission

Branches from LIP recovery. One flight can take only one branch.

- LIP / Back to mission -> Complete
- LIP / Back to mission -> loiter
- LIP / Back to mission -> POSCTL
- LIP / Back to mission -> EFL
- LIP / Back to mission -> Land

16. EFL descent

Branches from EFL descending. One flight can take only one branch.

- EFL / Descending -> Complete
- EFL / Descending -> loiter
- EFL / Descending -> POSCTL
- EFL / Descending -> EFL
- EFL / Descending -> Land

17. EFL loiter

Branches from EFL loiter. One flight can take only one branch.

- EFL / Loiter -> mission
- EFL / Loiter -> POSCTL
- EFL / Loiter -> EFL
- EFL / Loiter -> Land

18. EFL back to mission

Branches from EFL recovery. One flight can take only one branch.

- EFL / Back to mission -> Complete
- EFL / Back to mission -> loiter
- EFL / Back to mission -> POSCTL
- EFL / Back to mission -> EFL
- EFL / Back to mission -> Land

APPROVAL

Please mark a decision, sign, and return this page.

- ☐ Approved - use the 18-mission grouping as the official suite
- ☐ Approved - use only missions 1-8 (single-path, 18 original tests)
- ☐ Not approved - keep the full 77 Task / Element / Operation tests
- ☐ Approved with comments (write below)

Comments

Name

Role

Date

Signature
